

SMALLER LOOKS ROUNDER: CORRECTING RESOLUTION-DEPENDENT BIAS OF SHAPE DESCRIPTORS IN BIOIMAGE AND RADIOMICS SOFTWARE

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ABSTRACT

Circularity and sphericity are ubiquitous in cell morphometry and radiomics, but the perimeter and surface estimates behind them depend on resolution. The default estimators of scikit-image (hence CellProfiler), OpenCV and ImageJ make the same object rounder the fewer pixels it covers, which we explain with a closed-form model; IBSI sphericity of a sphere drops from 0.92 to 0.82 as CT slices thicken from 1 to 5 mm. Even Crofton estimators fail for small objects and thick slices, as they miss chords shorter than the sampling step. We show that the missed chords number one sixth of the single-sample runs, yielding a parameter-free run-corrected Crofton estimator for 2D and anisotropic 3D grids. Averaged over shapes it is unbiased to within 0.005 down to 2-pixel radii, keeps smooth lesions' sphericity bias within 0.005 up to 5 mm slices, and is five times more accurate than PyRadiomics on lung-tumour phantoms. On re-segmented nuclei, the default estimator shifts the form factor by up to 3.3 standard deviations and reverses a biological contrast at $10\times$; corrected estimators do not.

Index Terms— Morphometry, radiomics, harmonization, digital geometry, stereology

1. INTRODUCTION

Shape descriptors of segmented objects are a staple of quantitative imaging. Nuclear circularity quantifies pleomorphism in histopathology, CellProfiler's *FormFactor* is part of every Cell Painting profile [1, 2], and sphericity is a standardized radiomics feature of the Image Biomarker Standardisation Initiative (IBSI) [3]. Both are ratios of a size to a boundary measure, $FF = 4\pi A/P^2$ and $\Psi = (36\pi V^2)^{1/3}/S$, equal to 1 for a disk and a ball. Studies routinely pool data of different resolution: slides scanned at $20\times$ and $40\times$, cell types of different sizes, CT series with 1–5 mm slices.

The area or volume of a mask is easy to estimate, since the pixel count is unbiased under random placement; the boundary is not [4, 5]. Kulpa [4] derived the large-object length factor of chain codes; Crofton estimators, which count boundary crossings of lattice lines, are unbiased in the large-object limit [6], and a recent comparison recommends them for 2D perimeters [7]; MATLAB patched its circularity for small ob-

jects in 2023 [8]. Three gaps remain: (a) no model explains the *size dependence* of the estimators shipped in open-source tools, which is what breaks comparisons across resolutions; (b) Crofton estimators are themselves biased when objects are small or slices thick, because chords shorter than the sampling step are never seen, and no correction exists; (c) the consequences in real pipelines are unquantified.

Contributions. (i) A *run-corrected Crofton* (RCC) estimator: we show that, for smooth boundaries, the expected number of missed chords equals one sixth of the number of single-sample runs, which removes the small-object and thick-slice bias without parameters, in 2D and on anisotropic 3D grids. (ii) A closed-form model, $P \approx \kappa(L - 2\pi\delta)$, of the default estimators' size dependence, and a post-hoc correction (F1) for existing feature tables. (iii) An audit of eight implementations against exact ground truth, and evidence on 89k real nuclei and 64 lung tumours of shifted features, distorted biological contrasts and degraded cross-resolution classification; IBSI's recommended isotropic resampling makes sphericity worse.

2. WHY DEFAULT ESTIMATORS DEPEND ON SIZE

Let $X \subset \mathbb{R}^2$ have a smooth boundary of length L , digitized on a unit grid as $D = X \cap \mathbb{Z}^2$ with random sub-pixel position and orientation. Most tools trace a polygon along D whose edges take a few lattice directions (Fig. 1a). If a boundary element ds at tangent angle θ becomes polygon length $g(\theta) ds$, and the polygon lies on average a distance δ inside ∂X , Steiner's formula (a closed curve moved inward by δ has length $L - 2\pi\delta$) gives

$$P \approx \kappa(L - 2\pi\delta), \quad \kappa = \frac{1}{2\pi} \int_0^{2\pi} g(\theta) d\theta. \quad (1)$$

Chain codes, used by scikit-image's default `regionprops` perimeter (hence CellProfiler), OpenCV's `arcLength` and MATLAB up to R2022b, join the centres of boundary pixels by steps of length 1 or $\sqrt{2}$: $g(\theta) = |\cos \theta| + (\sqrt{2} - 1)|\sin \theta|$ for $|\theta| \leq \pi/4$, so $\kappa = 8(\sqrt{2} - 1)/\pi \approx 1.055$ [4]. Along the grid axis closest to the normal, the last pixel centre inside X lies at a uniform distance in $[0, 1)$ from ∂X , i.e. $\cos \theta/2$ on average perpendicular to it, so $\delta = \frac{4}{\pi} \int_0^{\pi/4} \frac{\cos \theta}{2} d\theta = \sqrt{2}/\pi \approx$

0.45. For a disk of radius r pixels,

$$\text{FF}_{\text{chain}}(r) \approx \kappa^{-2} \left(r / (r - \sqrt{2}/\pi) \right)^2, \quad (2)$$

which exceeds 1 for $r < 8.7$ px and tends to 0.899. *Marching squares* (PyRadiomics; IBSI meshes) have the same κ but $\delta = 0$, a constant factor κ^{-2} on FF; *pixel-edge polygons* have $\kappa = 4/\pi$, $\delta = 0$; ImageJ cuts the corners of pixel-edge outlines, a hybrid; MATLAB R2023a corrects δ but not κ [8]. In 3D, marching cubes on binary masks overestimates the surface increasingly with the slice-to-pixel ratio (Fig. 1e). *FI (post hoc)*. Inverting Eq. (1), $\hat{P} = P_{\text{chain}}/\kappa + 2\pi\delta'$ repairs an existing scikit-image/CellProfiler perimeter column without the masks. Small objects (2–3 px) lose extra length, so we calibrate δ' once on the synthetic shapes (minimax FF error, giving 0.52) and use $\delta' = \frac{1}{2}$, i.e. $\hat{P} = P/1.055 + \pi$; no real data are used.

3. RUN-CORRECTED CROFTON ESTIMATOR

Crofton estimator. Consider lattice directions d (4 in 2D, 13 in 3D, in physical units for voxel spacing s , volume v). Lines along d through the pixel or voxel centres have density $\lambda_d = \|d\|/v$ per unit length or area orthogonal to d , and each line is sampled every $h_d = \|d\|$. If N_d is the number of runs (maximal sequences of object samples) along such lines, then N_d/λ_d estimates the object’s projection orthogonal to d (half the crossing measure for non-convex objects), and the Cauchy–Crofton formulas give

$$\hat{P} = \pi \sum_d w_d N_d / \lambda_d \quad (2\text{D}), \quad \hat{S} = 4 \sum_d w_d N_d / \lambda_d \quad (3\text{D}), \quad (3)$$

with w_d the fraction of directions (circle or sphere) closest to $\pm d$, computed for the actual spacing [6, 9].

Missed chords. A run is seen only if its chord contains a sample. With a uniformly random sample offset, a chord of length ℓ yields k samples with $P(k=0) = \max(0, 1 - \ell/h)$, and $P(k=1) = \ell/h$ for $\ell \leq h$, $2 - \ell/h$ for $h < \ell \leq 2h$. Short chords occur where a line grazes the boundary: at distance t from the tangent line of a boundary with curvature radius ρ , $\ell \approx 2\sqrt{2\rho t}$, and since t is uniform, the chord-length density is linear near zero, $f(\ell) \approx a\ell$. Hence the expected numbers of missed chords and of single-sample runs are

$$\begin{aligned} \mathbb{E}[\text{missed}_d] &= \int_0^h a\ell \left(1 - \frac{\ell}{h}\right) d\ell = \frac{ah^2}{6}, \\ \mathbb{E}[n_{1,d}] &= \int_0^h a\ell \frac{\ell}{h} d\ell + \int_h^{2h} a\ell \left(2 - \frac{\ell}{h}\right) d\ell = ah^2, \end{aligned} \quad (4)$$

so, whatever a and h , the missed chords are one sixth of the single-sample runs:

$$N_d^{\text{RCC}} = N_d + \frac{1}{6} n_{1,d}, \quad (5)$$

where $n_{1,d}$ counts runs of exactly one sample along d . Plugging N_d^{RCC} into Eq. (3) gives the RCC estimator. It needs no

parameter, no resampling and one extra comparison per direction; on anisotropic grids h_d differs between directions, which Eq. (5) handles automatically. A quadratic density $a\ell + b\ell^2$ gives the second-order variant $(19n_1 - 4n_2)/66$, useful for irregular boundaries.

4. EXPERIMENTS

Synthetic ground truth. 5,000 shapes of known perimeter and area: ellipses (axis ratios 1, 0.6, 0.35) and two families of random smooth star-shaped “nuclei”, equivalent radii 2–48 px, random rotation and sub-pixel offset. In 3D, randomly rotated spheres and 1:0.75:0.5 ellipsoids (equivalent radius 3–20 mm) on $1 \times 1 \times s$ mm grids, $s \in [1, 5]$. We run the libraries’ own code (scikit-image 0.26 [10], OpenCV 4.13, PyRadiomics 3.0.1 [11]); CellProfiler’s FormFactor uses scikit-image’s perimeter (checked in its source); ImageJ’s [12] traced perimeter is ported, MATLAB R2023a’s formula [8] implemented.

Real nuclei. PanNuke fold 1 [13] (H&E, $40\times$, annotated types; 43,519 nuclei) and BBBC038 [14] (22,858 nuclei). Lower magnification is obtained by *re-digitizing* each annotated mask on a $2\times$ or $4\times$ coarser grid (block mean ≥ 0.5), and, realistically, by *re-segmenting* block-averaged images independently at each resolution with the pretrained Cellpose [15] `nuclei` model, matching nuclei by IoU ≥ 0.5 (12,279 PanNuke and 10,454 BBBC038 nuclei). As the true shape is unknown, we report the paired shift Δ/σ (mean change in FF over its native s.d.; bootstrap s.e. ≈ 0.01), Spearman’s ρ_A between area and FF, Cohen’s d of FF between PanNuke nucleus types, and the balanced-accuracy drop at $10\times$ of a $40\times$ -trained logistic regression predicting the type from area, solidity, eccentricity, extent, axis lengths and FF (5 image-level splits).

Lung tumours. The 63 CTs of the Medical Segmentation Decathlon lung set [16] provide real tumour shapes; as their slice-wise annotations change when re-digitized, each tumour ($\geq 300 \text{ mm}^3$; 64 tumours, radius 3–45 mm) is turned into a phantom: the mask is resampled to 0.5 mm and smoothed ($\sigma = 1.5 \text{ mm}$), its 0.5 level set defines a surface of known sphericity (0.60–0.96, from a fine mesh), point-sampled at 0.7 mm in-plane with 0.7–5 mm slices.

5. RESULTS

Synthetic 2D (Fig. 1b). The chain-code tools overestimate circularity by +0.55 at $r = 2$ px and +0.11 at $r = 4$ px and underestimate it by 0.08 at $r = 48$, as Eq. (2) predicts (dashed). ImageJ follows the same trend (+0.09 to -0.08); PyRadiomics and MATLAB R2023a are offset by κ^{-2} (-0.08 to -0.09). Crofton is unbiased for large objects but overestimates circularity by 0.050 at $r = 2$ and 0.013 at $r = 4$ px. RCC reduces this to ≤ 0.005 at every size, at

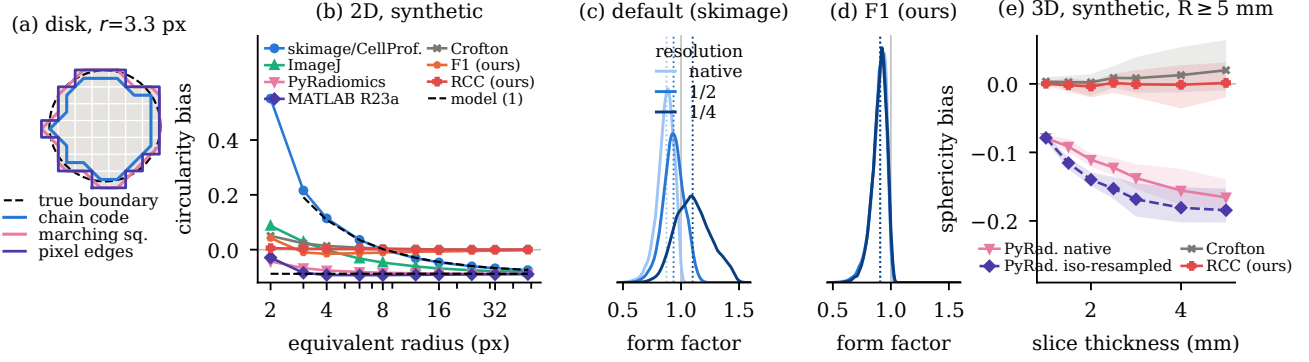


Fig. 1. (a) Polygons traced on a digitized disk: chain codes run inside the boundary ($\delta > 0$), marching squares on it, pixel edges outside. (b) Mean circularity bias on 5,000 synthetic shapes vs. size; dashed: model, Eq. (1). (c,d) FF of 10,454 BBBC038 nuclei segmented independently by Cellpose at native, $\frac{1}{2}$ and $\frac{1}{4}$ resolution (dotted: medians), with the default estimator (c) and F1 (d). (e) Sphericity bias of randomly rotated spheres/ellipsoids ($R \geq 5$ mm) vs. slice thickness; bands: 10–90% range.

Table 1. Real nuclei: paired shift Δ/σ of FF when the resolution is halved ($\frac{1}{2}$) or quartered ($\frac{1}{4}$), by re-digitization of the annotations (“re-dig.”) or independent Cellpose re-segmentation (“re-seg.”), and area–FF correlation ρ_A at native resolution. Ideal: zero shift.

FF from	PanNuke (H&E)					BBBC038 (fluorescence/brightfield)				
	re-dig.		re-seg.		ρ_A	re-dig.		re-seg.		ρ_A
	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$		$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$	
scikit-image / CellProfiler	+0.72	+3.25	+0.68	+2.53	−0.34	+0.92	+5.56	+0.95	+3.30	−0.56
ImageJ	+0.34	+0.96	+0.31	+0.96	−0.21	+0.41	+1.31	+0.48	+1.68	−0.41
MATLAB R2023a	+0.06	+0.27	0.00	+0.16	−0.05	+0.11	+0.51	+0.04	+0.20	−0.19
Crofton	+0.11	+0.31	+0.06	+0.25	−0.09	+0.16	+0.51	+0.13	+0.46	−0.24
F1 (ours, post hoc)	+0.03	+0.16	−0.03	+0.05	−0.03	+0.07	+0.35	−0.01	+0.05	−0.16
RCC (ours)	+0.09	+0.10	+0.03	+0.09	−0.08	+0.11	+0.12	+0.10	+0.28	−0.22

Table 2. PanNuke: Cohen’s d of FF between nucleus types (inflammatory, connective vs. neoplastic; s.e. ≈ 0.014), within one magnification and with inflammatory nuclei at $20\times$ vs. neoplastic at $40\times$ (“mixed”); balanced-accuracy drop of a $40\times$ -trained type classifier at $10\times$ (no FF: 0.008).

FF from	infl.–neo.		conn.–neo.		mixed	drop
	$40\times$	$10\times$	$40\times$	$10\times$		
scikit-image / CellProfiler	0.67	0.32	−0.62	+0.11	1.62	0.115
OpenCV	0.67	0.82	−0.62	−0.01	1.61	0.106
ImageJ	0.52	0.67	−0.71	−0.48	1.05	0.064
PyRadiomics, Crofton	0.40	0.53	−0.76	−0.62	0.56	0.041
F1 (ours)	0.35	0.39	−0.79	−0.41	0.35	0.054
RCC (ours)	0.38	0.31	−0.77	−0.72	0.49	0.031

similar variance; for the most irregular shapes it halves the bias at $r = 2$ ($0.10 \rightarrow 0.05$; second order: 0.03).

Synthetic 3D (Fig. 1e). PyRadiomics sphericity is biased by -0.08 at 1 mm and -0.17 at 5 mm slices for every lesion size; isotropic resampling, as IBSI recommends, *increases* the bias to -0.19 . Crofton drifts to $+0.046$ for $R = 5$ mm at 5 mm slices; RCC stays within ± 0.005 for all $R \geq 5$ mm and

all thicknesses. On the 64 (irregular) lung-tumour phantoms, PyRadiomics is biased by -0.065 at 0.7 mm and -0.136 at 5 mm slices, Crofton by $+0.005$ and $+0.042$, and RCC by $+0.002$ and $+0.026$ (10–90%: $[0.004, 0.050]$), a fivefold smaller error than PyRadiomics; the residual comes from tumour poles thinner than a slice and from boundary irregularity. Re-digitizing the raw annotations instead raises the mean sphericity at 5 mm for every estimator (PyRadiomics $0.57 \rightarrow 0.66$), because thick slices remove genuine slice-wise contour jitter.

Real nuclei (Fig. 1c,d; Table 1). With the default estimator, halving the resolution shifts the mean FF by $0.7\text{--}0.95\sigma$ and quartering it by $2.5\text{--}5.6\sigma$ in both settings and modalities; at $\frac{1}{4}$ resolution the median BBBC038 nucleus has an impossible FF > 1 . The default also creates an area–roundness correlation ($\rho_A = -0.34, -0.56$). Crofton reduces the shifts to $0.06\text{--}0.51\sigma$ and RCC to $0.03\text{--}0.28\sigma$; F1, whose calibrated offset may also absorb part of the smoothing of low-resolution masks, is the most stable under re-segmentation ($\leq 0.05\sigma$). Residual shifts are partly genuine: coarse grids remove real boundary detail.

Biological contrasts (Table 2). Inflammatory nuclei are

smaller than neoplastic ones (median radius 9.5 vs. 13.9 px at 40 $\times$), so the default inflates their extra roundness from $d = 0.35\text{--}0.40$ to 0.67. At 10 $\times$, the default reverses the sign of the connective–neoplastic contrast ($-0.62 \rightarrow +0.11$), whereas RCC preserves it ($-0.77 \rightarrow -0.72$). When scanners are mixed (inflammatory at 20 $\times$, neoplastic at 40 $\times$), the default reports $d = 1.62$ instead of 0.35–0.49. A 40 $\times$ -trained type classifier loses 0.115 balanced accuracy at 10 $\times$ with the default FF and 0.031 with RCC.

6. DISCUSSION

The size dependence of the default circularity of scikit-image, CellProfiler and OpenCV is predictable (Eq. (2)) and distorts any comparison across pixel sizes; constant offsets (PyRadiomics, MATLAB R2023a) mislead only absolute thresholds. RCC removes the residual sampling bias of Crofton estimators at negligible cost. We recommend RCC for new analyses on the native grid, F1 to repair existing CellProfiler/scikit-image tables, and reporting object sizes in pixels. Limitations: Eq. (5) assumes smooth boundaries (objects of about one slice remain biased), and block-averaging only approximates re-imaging. Code: <https://github.com/Hannibal0319/isbi2027-shape-bias>.

7. COMPLIANCE WITH ETHICAL STANDARDS

Only public, anonymized datasets were used; no ethical approval was required.

8. ACKNOWLEDGMENTS

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